

Electronics & Communication Systems

Course Introduction



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Pietro Nannipieri

- Digital Systems Architectures, VHDL
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 Nome ▾	Data/ora modifica ▾	Modificato da ▾	+ Aggiungi colonna ▾
  Lecture Slides	Circa un minuto fa	Pietro Nannipieri	
  VHDL & Hands-on Exercises	Circa un minuto fa	Pietro Nannipieri	

Course Material

Slides, VHDL Code from exercise, Lesson recordings

Tools

PC-based VHDL simulation tools:

- Aldec (<http://www.aldec.com>): Active HDL
- https://www.aldec.com/en/products/fpga_simulation/active_hdl_student
- **Mentor** (<http://www.mentor.com>): ModelSim
- <http://ghdl.free.fr/> (+ <http://gtkwave.sourceforge.net/>)

Xilinx Development tool:

- <http://www.xilinx.com/products/design-tools/vivado/vivado-webpack.html>



OVERVIEW

ModelSim

ModelSim simulates behavioral, RTL, and gate-level code - delivering increased design quality and debug productivity with platform-independent compile. Single Kernel Simulator technology enables transparent mixing of VHDL and Verilog in one design.

Watch webinar

View fact sheet

Get in touch with our sales team 1-800-547-3000



Tools – VHDL Simulation

<https://eda.sw.siemens.com/en-US/ic/modelsim/>

Student edition Currently not available

Questa*-Intel® FPGAs Pro Edition Software Version 24.2

ID	Date	Software Type	Software Package	Version	Operating Systems
826872	7/8/2024	Simulation Tools	Questa*-Intel® FPGAs Pro	24.2 (Latest)	Windows, Linux

The Questa*-Intel® FPGA Pro Edition, Version 24.2 includes functional and security updates. Users should keep their software up-to-date and follow the [technical recommendations](#) to help improve security. Additional security updates are planned and will be provided as they become available. Users should promptly install the latest version upon release.

The Questa*-Intel® FPGA Pro Edition, Version 24.2 is subject to removal from the web when support for all devices in this release are available in a newer version, or all devices supported by this version are obsolete. If you would like to receive customer notifications by e-mail, please subscribe to our [subscribe to our customer notification mailing list](#).

Downloads

[Linux Software](#)

[Windows Software](#)

Questa Software

Questa*-Intel® FPGA and Starter Editions

Download
QuestaSetup-24.2.0.40-windows.exe

Size: 2.2 GB

SHA1: 9d67fbaee4f76d81d550684993a16ea816858511

** Installation size: 9.83 GB

Tools – VHDL Simulation

[LINK](#) 2024 Windows

[LINK](#) 2024 Unix

[LINK](#) 2020 Windows
(smaller installation size)

[LINK](#) 2020 Unix
(smaller installation size)

Tool We will use in class

ModelSim-Intel® FPGA
Edition (includes Starter
Edition)

2020 suggested edition
(plug-in license)

Modelsim

The screenshot shows the Intel FPGA Software Download Center. The search bar at the top left contains the text "quartus lite 20" and is highlighted with a red box. Below the search bar, the left sidebar shows filter options: "Intel® Quartus Software", "Intel® Quartus® Prime Design Software (4)", "Additional Software", "Intel® FPGA Programming Software (4)", and "Intel® FPGA Simulation Tools (4)". The main content area displays "4 Results" and a list of software packages. The first result is "Intel® Quartus® Prime Lite Edition Design Software Version 20.1.1 for Linux". The second result is "Intel® Quartus® Prime Lite Edition Design Software Version 20.1.1 for Windows". A red arrow points from the second result to a detailed download page for "ModelSim-Intel® FPGA Edition (includes Starter Edition)". This page shows two download options: "ModelSimSetup-20.1.1.720-windows.exe" (Size: 1.2 GB, SHA1: d484e4c7) and "QuartusLiteSetup-20.1.1.720-windows.exe" (Size: 1.6 GB, SHA1: 5edd76fc). Below the download options, there are two footnotes: "** Nios® II EDS on Windows requires Ubuntu 18.04 LTS on Windows Subsystem for Linux" and "** Nios® II EDS requires you to install an Eclipse IDE manually."

<https://fpgasoftware.intel.com/>

PRODUCTS

FPGA Simulation

Active-HDL

Functional Verification

Riviera-PRO

ALINT-PRO

Emulation & Prototyping

HES-DVM

HES Proto-AXI

HES Boards

TySOM Boards

Daughter Cards

RTAX/RTSX Adaptor Boards

RTAX/RTSX Netlist Converter

Cloud

HES-DVM Proto Cloud Edition

Embedded

TySOM™ EDK

Requirements Management

Spec-TRACER

Mil/Aero Verification

DO-254/CTS

University Programs

VIP/IP Products

Active-HDL™ | Student Edition

Free Download

Free Active-HDL Student Edition

Active-HDL Student Edition is a mixed language design entry and simulation tool offered at no cost by Aldec for students to use during their course work.

Licensing

Active-HDL Student Edition includes a "load and go" license. This means students can begin using it immediately after installing.

Key Features of Active-HDL Student Edition

- Mixed language simulator
- Multi-FPGA & EDA Tool Design Flow Manager
- Graphical Design entry & editing
- Code2Graphics and Graphics2Code
- Pre-compiled FPGA vendor libraries
- IEEE Language Support: VHDL, Verilog, SystemVerilog(Design), SystemC
- Waveform Viewer and List Viewer
- Interface with MATLAB®/Simulink®
- HTML and PDF Design Documentation

Tools – VHDL Simulation

- [LINK](#)
- Low Cost Alternative Solution

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Tools

VHDL Logic synthesis and FPGA Programming ([LINK](#))

Vivado

Downloads

[Log4j Vulnerability Patch](#)[Licensing Help](#)[Alveo](#)

[Vivado \(HW Developer\)](#)[Vitis \(SW Developer\)](#)[Vitis Embedded Platforms](#)[Alveo Packages](#)[PetaLinux](#)

Version	We strongly recommend using the latest releases available.
2021.2	2020
2021.1	2020.2
2020.2	
Vivado Archive	
CAE Vendor Libraries Archive	

Vivado Design Suite - HLx Editions - 2020.2 Full Product Installation

[Important Information](#)[Download](#)



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<https://www.xilinx.com/support/download.html>

Vivado



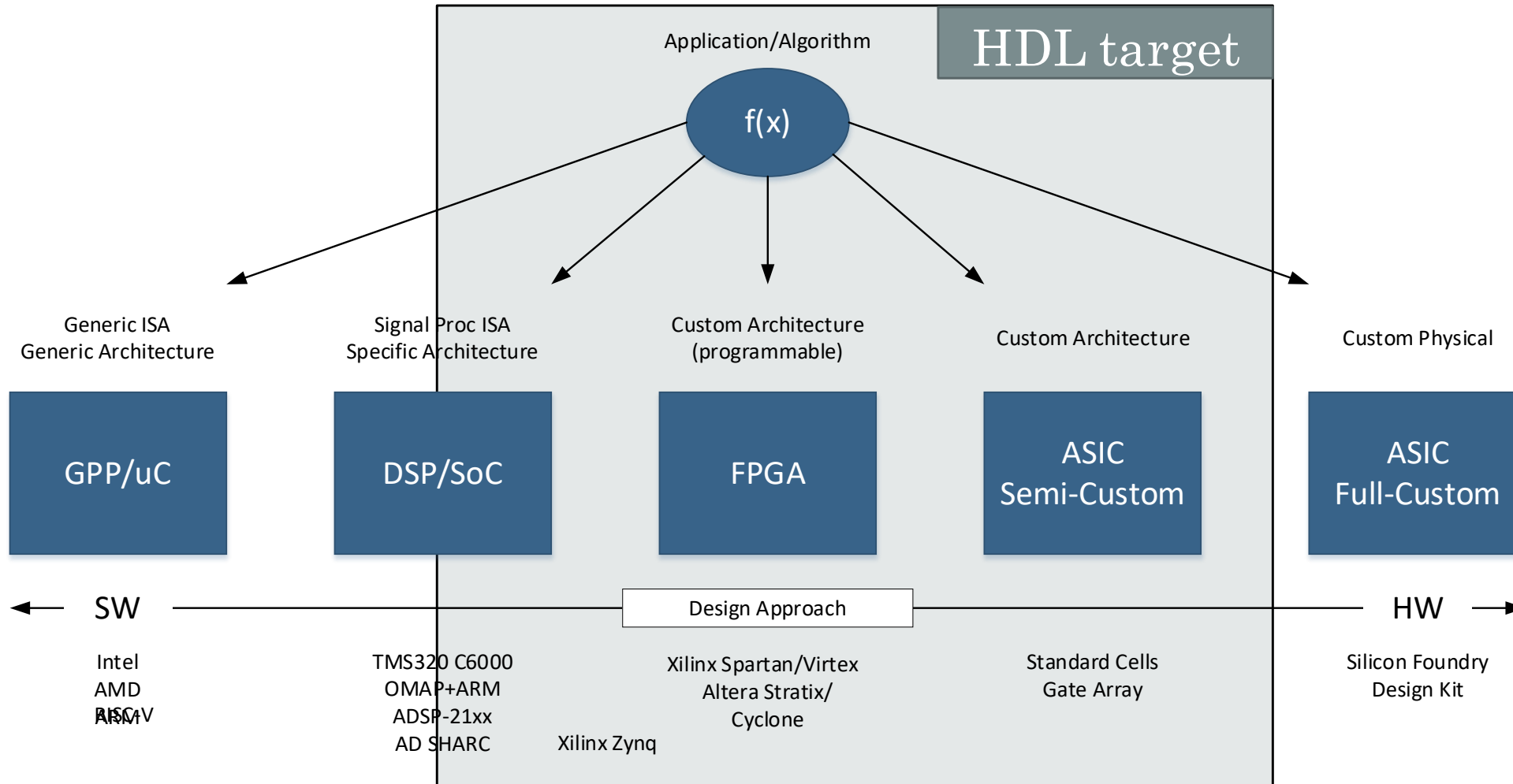
Free Version

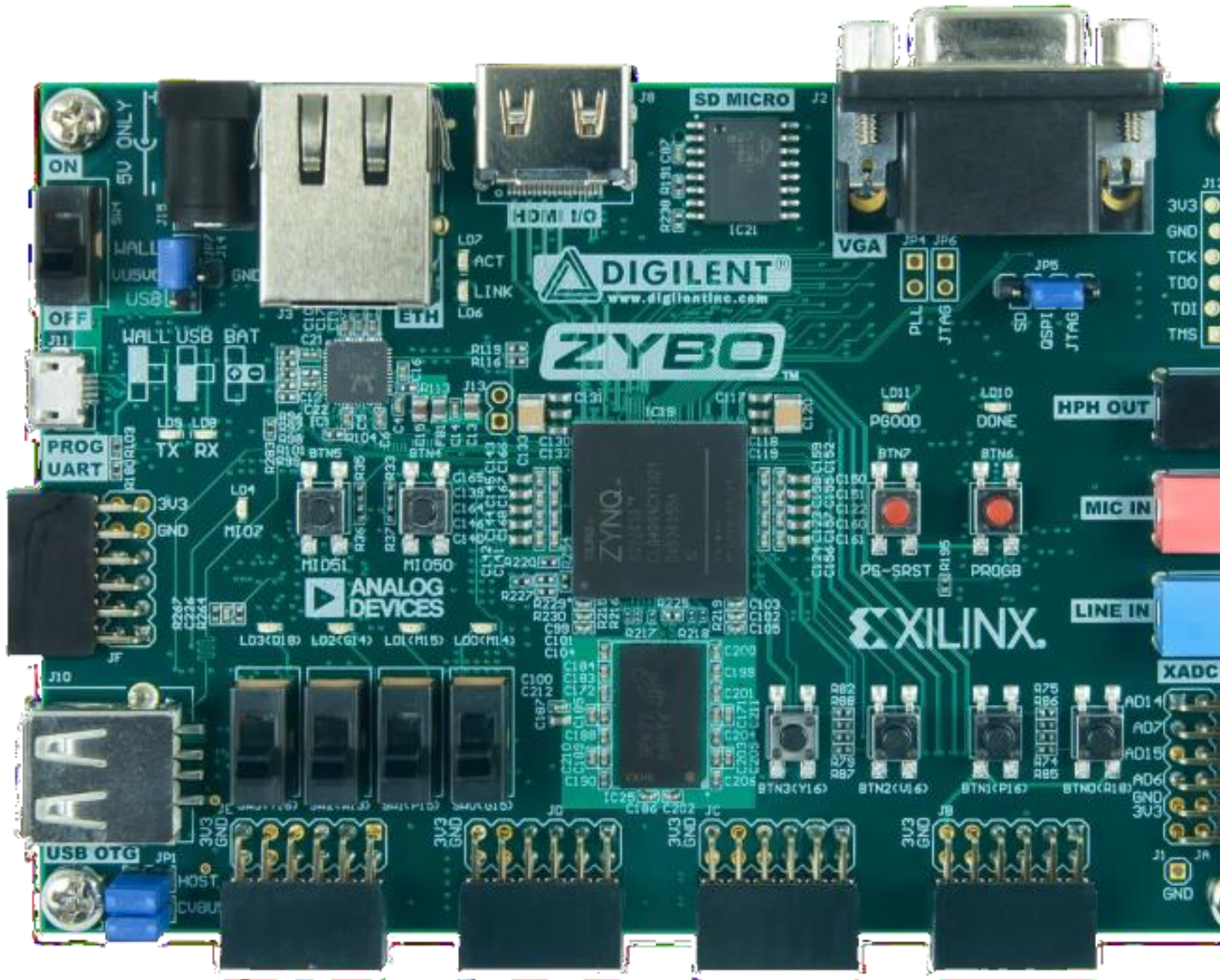


To save some space

Digital Design Technology Map:

From a $f(x)$ to *Hardware Acceleration*





Hands-on:

VHDL modelling, logic
synthesis and FPGA
Programming

During exercitation we will
design from scratch a
complete digital system, from
requirements to the
implementation on a FPGA
Board and its test.

Hands-on: DDFS System Description

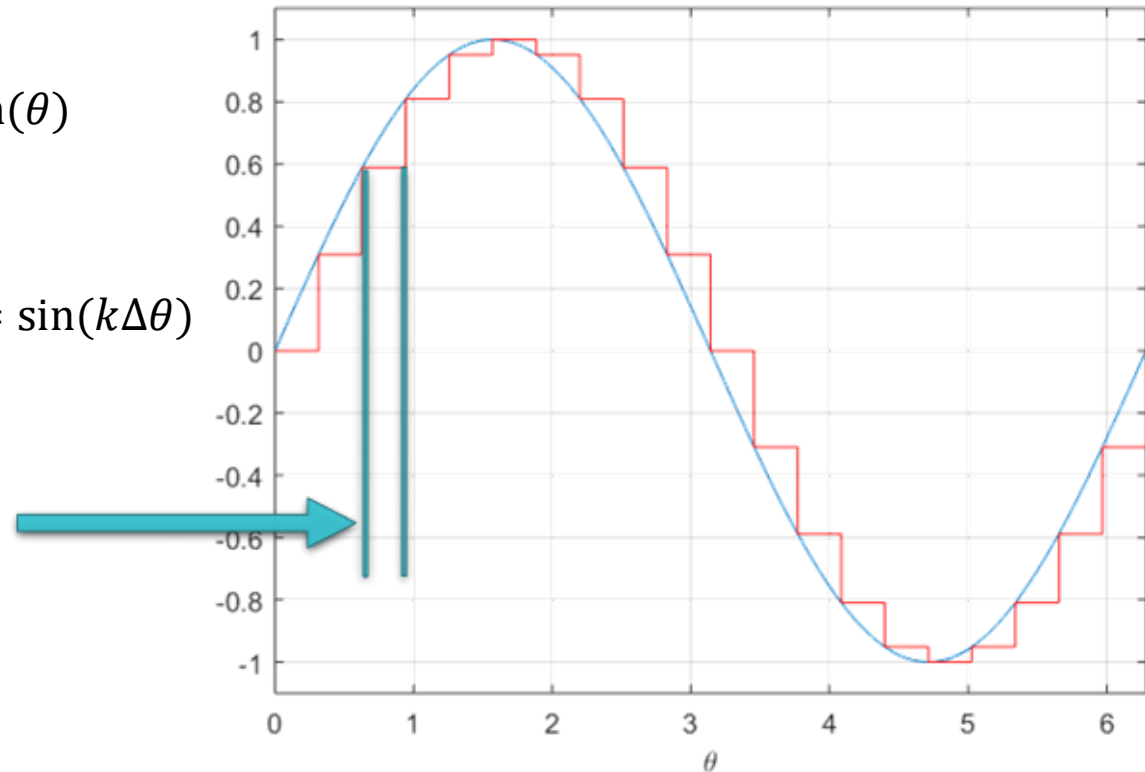
The DDFS (Direct Digital Frequency Synthesizer) is a digital system able to generate a quantized *sin* waveform, with frequency control.

Building block of many Telecommunication systems

Suppose we want to generate a single tone at frequency f :

- Ideal output: $\sin(2\pi f t) = \sin(\theta)$
- Discrete time: $\sin(2\pi f k T_{ck}) = \sin(k\Delta\theta)$
- Phase resolution:
 - Depends on f
 - Depends on f_{ck}

$$\Delta\theta = 2\pi \frac{f}{f_{ck}}$$



Exam

- Oral test on topics covered in the course
- Digital Design Project (Optional BUT Recommended)
 - (VHDL and automatic logic synthesis for FPGA technology)
 - Realization of digital electronic system of similar complexity as seen during class hands-on exercises
 - 1 - 2 people
 - Request for project assignment via email
 - Email delivery of source code and testbench together with the technical report
 - Design Review/Discussion before the oral exam

Questions?

